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Jiangcheng You

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Education

University of Science and Technology of China , Hefei, Anhui, China *Sept. 2022 – Present*
Ph.D. in Mathematics.
Supervisor: Hao Yin.

Southeast University , Nanjing, Jiangsu, China *Sept. 2016 – June 2020*
B.S. in Mathematics.

Research Interests

- Scalar curvature and Riemannian geometry: rigidity, stability, and topological constraints.
- Geometric relativity: positive mass, asymptotically flat and ALE geometry, and geometric problems motivated by general relativity.
- Yang–Mills theory and gauge theory: connections, moduli spaces, and geometric partial differential equations.

Research Works

- Shuchen Liu, **Jiangcheng You** and Junbiao Zhou, *Harmonic Coarse Left Inverses for Quasi-Isometric Embeddings of the Hyperbolic Plane*. Preprint, 2026.
- **Jiangcheng You** and Heng Zhang, *Area Nondecreasing Drawstrings and Counterexamples to Euclidean Rigidity and Zero Mass Stability*. Preprint, 2026. hexagon:2610.00435v1.
- **Jiangcheng You** and Heng Zhang, *Immersed Hypersurfaces with Positive First Newton Transformation*. Preprint, 2026. arXiv:2609.26665.
- **Jiangcheng You** and Heng Zhang, *Topological Rigidity of Contractible* 3*-Manifolds and Handlebody Interiors under Nonnegative Scalar Curvature*. Preprint, 2026. arXiv:2607.25015.
- **Jiangcheng You**, *Intrinsic Brown–York Type Mass at Infinity in Four Dimensions*. Preprint, 2026. arXiv:2607.02399.
- **Jiangcheng You** and Heng Zhang, *Reciprocal Sums of Neumann Eigenvalues in Non-Euclidean Space Forms*. Preprint, 2026. arXiv:2606.27848.
- **Jiangcheng You**, *Curvature at Infinity of Scalar-Flat ALE Four-Manifolds*. Preprint, 2026. arXiv:2606.16176.
- **Jiangcheng You** and Heng Zhang, *Gromov's Euclidean Endpoint* C^0 *Rigidity for the Positive Mass Theorem*. Preprint, 2026. arXiv:2606.15372.

Academic Talks

- **Green Functions and Rigidity under Nonnegative Scalar Curvature**, Geometry and Topology Seminar, Institute of Mathematics, AMSS, Chinese Academy of Sciences, September 10, 2026.
- **Gromov's Euclidean Endpoint** C^0 **Rigidity for the Positive Mass Theorem**, Geometry and Topology Seminar, online, June 30, 2026.
  Organized by Lizhi Chen (Lanzhou University) and Bo Zhu (YMSC, Tsinghua University).
- **Curvature at Infinity of Scalar-Flat ALE Four-Manifolds**, NJU–SJTU Geometric Analysis Seminar, Nanjing University, May 30, 2026.

Academic Visits

- Institute of Mathematics, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China, September 9–12, 2026.

Scholarships

- Second Class Academic Scholarship for Master's Students, University of Science and Technology of China, 2022, 2023.
- Second Class Academic Scholarship for Ph.D. Students, University of Science and Technology of China, 2024, 2025, 2026.